

Axel Delakowski

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SKILLS

Programming: Python, Java, JavaScript, TypeScript, Node.js

Machine Learning/AI: Mediapipe, Vertex AI, TensorFlow, PyTorch, OpenRouter, LangGraph

Frameworks/Tools: Spring Boot, React, Next.js, Docker, Google Cloud Platform, Git

Web/Databases: ClickHouse, SQLite, Python FastAPI, SQL (MySQL, Azure SQL), Twilio

EXPERIENCE

Product Engineer (Software Focus), OrganOx

December 2024 - Present

- Developing automated data ingestion and analysis pipelines using Python, replacing manual workflows with 10 validated command-line tools to ensure high-fidelity data processing for medical telemetry, accelerating project by 3 months
- Led software validation initiatives and scaled a rigorous testing framework (Unit/Integration) to audit embedded packages, enforcing strict data integrity standards and optimizing production code deployment velocity.
- Optimizing liver support system control loop algorithms by refactoring legacy firmware tech debt and implementing rigorous validation protocols to ensure precise, real-time physiological response handling.

Founding Engineer, HealthcareAIGuy, Penn Venture Labs

December 2024 - Present

- Engineered a Secure Member Directory Portal by implementing LinkedIn OAuth and Java Spring Boot middleware to proxy Airtable API interactions, shielding sensitive keys from the frontend.
- Architected an automated data harvesting pipeline utilizing Python, BeautifulSoup4, and Selenium to orchestrate weekly headless-crawling of unstructured newsletter data, transforming raw text into a structured venture capital database.
- Deployed containerized batch-processing jobs on GCP Cloud Run via Docker and GitHub Actions, utilizing Cloud Scheduler, Cloud Build, and Secret Manager to achieve hands-off production execution and strict data deduplication.
- Built a granular LLM testing framework to validate model logic while minimizing API expenditure, driving a 50% increase in MRR for a newsletter platform.

Software Engineer Intern, Airalux Medical

September 2024 - December 2024

- Engineered a scalable full-stack solution (Swift/iOS, Python) for a medical device reducing post-op pneumonia risk, facilitating a 3x increase in breathing exercises and statistically significant reductions in hospital length of stay.
- Established a HIPAA-compliant data pipeline by migrating production data from Airtable to Azure SQL, securing a stable relational architecture for high-volume medical device telemetry.

Deep Learning Research Assistant, Intelligent Media Processing Laboratory

April 2020 - August 2020

- Architected parallel training pipelines and dashboards for a generative framework (PSIG-GAN ISSN: 2332-5615), accelerating model convergence by 300%.

EDUCATION/CERTIFICATIONS

University of Pennsylvania, Master of Applied Science: Computer Science - AI Concentration

May 2027

The College of New Jersey, Bachelor of Science: Biomedical Engineering

May 2021

- Mayo Business Plan Competition 1st place, translated Harvard research into a circular economy

PROJECTS

Javier: Autonomous NYC Rental Concierge | Lead Backend (Agentic Engineering Hack)

May 2026

- Decomposed an ambiguous, fragmented market problem to architect and deploy a multi-service agentic monorepo from scratch during an intense 5.5-hour sprint, demonstrating extreme technical ownership across ingestion, analytics, and execution layers.
- Built a high-throughput data pipeline using Nimble and SQLite synced to a ClickHouse analytics engine for real-time market simulation, feeding a closed-loop OpenRouter/Hono agent that autonomously calls tools to verify data, text brokers via Twilio, schedule tours, and execute web3 escrow via Coinbase CDP.

VisionLink | Lead Developer (MedGemma Impact Challenge)

December 2025 - February 2026

- Synthesized a decoupled "Relay Race" agentic pipeline using LangGraph to automate rural ophthalmic triage, leveraging PaliGemma (3B) for visual feature extraction and MedGemma (2B/4B) for clinical reasoning and patient interviewing.

Kinemax Engine | Lead Developer (Tennis Swing Analysis)

October 2025 - December 2025

- Architected a computer vision pipeline using FastAPI and Docker to extract 33-point 3D skeletal landmarks from 2D tennis footage via MediaPipe Pose for markerless motion capture.

HARP: A Hybrid Auto-RNN Prediction System (SPARC Hackathon)

June 2025 - August 2025

- Applied Echo State Networks (ESNs) and RNNs to create an accessible data analysis platform, democratizing advanced neural networks for professional researchers.

FUN FACTS

- Classically trained pianist (3x Carnegie Hall performer), Chinese & German speaker, and extensive global traveler